

# Thunder Hash Family

## Implementation Self-Assessment Report

x86-64 Architecture — Reference Implementation Version

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# 1 Assessment Scope and Material

This report records the implementation self-assessment of the Thunder cryptographic hash family for the Reference Implementation Version on the x86-64 Architecture. It follows the *Guidelines for Self-Assessment of Cryptographic Algorithms Performance on x86 Architecture*. The assessment covers fixed-output hash computation.

| Item                   | Description                                                     |
|------------------------|-----------------------------------------------------------------|
| Algorithm category     | Cryptographic hash algorithm                                    |
| Algorithm name         | Thunder Hash Family                                             |
| Sub-function           | Fixed-output hash computation                                   |
| Architecture           | x86-64                                                          |
| Implementation version | Reference Implementation Version                                |
| Material package       | Cryptographic Hash-Thunder-x86-Reference Implementation Version |
| Instances covered      | Thunder-512, Thunder-768, and Thunder-1024                      |

# 2 Algorithm Parameters

| Instance     | State (bits) | Digest (bits) | Preimage | Collision | Data limit    |
|--------------|--------------|---------------|----------|-----------|---------------|
| Thunder-512  | 1600         | 512           | 512      | 256       | $2^{64}$ bits |
| Thunder-768  | 1600         | 768           | 768      | 384       | $2^{64}$ bits |
| Thunder-1024 | 1600         | 1024          | 1024     | 512       | $2^{64}$ bits |

# 3 Assessment Environment and Build

The operating-system, kernel, machine, hostname, and timestamp fields are taken from the ngcc JSON evidence. Processor, memory, storage, compiler, and build-tool information is recorded from the machine commands used for the assessment.

| Component                    | Configuration                                                                                         |
|------------------------------|-------------------------------------------------------------------------------------------------------|
| Hostname                     | user-Super-Server                                                                                     |
| CPU model                    | Intel Xeon(R) Platinum 8358 CPU @ 2.60GHz (64 logical CPUs)                                           |
| Clock frequency              | 2.60 GHz nominal frequency                                                                            |
| Instruction set / machine    | x86-64 execution environment; AVX2-enabled implementation where applicable; ngcc machine field x86_64 |
| Cores used                   | Single core for reported measurements                                                                 |
| Memory / storage             | 256 GB ECC DDR4-3200 RAM; 1 TB storage                                                                |
| Operating system             | Ubuntu 22.04 LTS; kernel 6.8.0                                                                        |
| Kernel / release             | 6.8.0                                                                                                 |
| Compiler                     | GCC 11.4                                                                                              |
| Build tool                   | CMake 3.22.1                                                                                          |
| Third-party crypto libraries | None; the Thunder implementation is self-contained C code                                             |
| Run start / end time         | 2026-06-30T13:26:43+0800 /<br>2026-06-30T13:31:57+0800                                                |

**Build configuration.** Portable C reference implementation using the ISO C99-oriented reference build configuration.

```
gcc -std=c99 -Wpedantic -Wall -Wextra -O2
```

## 4 Test Method and Input Data

The assessment uses `ngcc` for functional, performance, memory, and stability testing. The script `testThunderX64.sh` is run from `ngcc_bench-main/build`; it invokes `ngcc_bench` with `-test hash -mode all`. For performance measurements, `ngcc` uses 10000 iterations per S-level input size, exceeding the guideline recommendation of no fewer than 100 measurements.

| Test item           | Input data and method                                                                                        |
|---------------------|--------------------------------------------------------------------------------------------------------------|
| Functional testing  | KAT files in the submitted test-vector directories for Thunder-512, Thunder-768, and Thunder-1024.           |
| Performance testing | S1–S8 fixed input lengths: 32, 128, 512, 1024, 4096, 8192, 16384, and 65536 Bytes.                           |
| Resource testing    | Static memory and peak memory reported by <code>ngcc</code> in Bytes.                                        |
| Randomness          | Not used for this hash-function assessment. The performance inputs and KAT files are fixed and reproducible. |

### KAT File Inventory

The complete KAT vector files are included in the corresponding **Self-Assessment Code and Data** package. The inventory below records the file names and exact byte sizes present in the submitted test-vector directories. The ARM and x86 packages contain matching copies.

| Instance     | KAT file           | Size (Bytes) |
|--------------|--------------------|--------------|
| Thunder-512  | KAT_2_12_short.txt | 2804822      |
|              | KAT_2_23_valid.txt | 815          |
|              | KAT_2_33_valid.txt | 824          |
|              | KAT_Loop_valid.txt | 2220         |
| Thunder-768  | KAT_2_12_short.txt | 3067030      |
|              | KAT_2_23_valid.txt | 1007         |
|              | KAT_2_33_valid.txt | 1016         |
|              | KAT_Loop_valid.txt | 2284         |
| Thunder-1024 | KAT_2_12_short.txt | 3333335      |
|              | KAT_2_23_valid.txt | 1202         |
|              | KAT_2_33_valid.txt | 1211         |
|              | KAT_Loop_valid.txt | 2349         |

## 5 Assessment Results

All results below are populated from the `ngcc` JSON files listed in Section 6.

### 5.1 Functional Testing

| Instance     | Test set                              | Result                     |
|--------------|---------------------------------------|----------------------------|
| Thunder-512  | All KAT files listed in the inventory | PASS (4104/4104, failed 0) |
| Thunder-768  | All KAT files listed in the inventory | PASS (4104/4104, failed 0) |
| Thunder-1024 | All KAT files listed in the inventory | PASS (4104/4104, failed 0) |

## 5.2 Performance Testing

| Instance     | Size         | Avg. time (ns) | Avg. cycles | Cycles/Byte | Throughput (MB/s) |
|--------------|--------------|----------------|-------------|-------------|-------------------|
| Thunder-512  | S1 (32 B)    | 1615.00        | 4112.42     | 128.51      | 19.21             |
|              | S2 (128 B)   | 3153.00        | 8113.83     | 63.39       | 39.95             |
|              | S3 (512 B)   | 7761.00        | 20093.20    | 39.24       | 65.54             |
|              | S4 (1024 B)  | 13889.00       | 36027.08    | 35.18       | 73.46             |
|              | S5 (4096 B)  | 50711.00       | 131763.62   | 32.17       | 80.69             |
|              | S6 (8192 B)  | 99787.00       | 259359.76   | 31.66       | 82.05             |
|              | S7 (16384 B) | 197834.00      | 514278.59   | 31.39       | 82.79             |
|              | S8 (65536 B) | 787074.00      | 2046296.45  | 31.22       | 83.26             |
| Thunder-768  | S1 (32 B)    | 1642.00        | 4184.09     | 130.75      | 18.91             |
|              | S2 (128 B)   | 3196.00        | 8226.12     | 64.27       | 39.42             |
|              | S3 (512 B)   | 9423.00        | 24416.88    | 47.69       | 54.04             |
|              | S4 (1024 B)  | 17197.00       | 44626.91    | 43.58       | 59.37             |
|              | S5 (4096 B)  | 66949.00       | 173982.28   | 42.48       | 61.13             |
|              | S6 (8192 B)  | 133879.00      | 347996.92   | 42.48       | 61.16             |
|              | S7 (16384 B) | 265999.00      | 691506.12   | 42.21       | 61.58             |
|              | S8 (65536 B) | 1062442.00     | 2762252.60  | 42.15       | 61.68             |
| Thunder-1024 | S1 (32 B)    | 1618.00        | 4124.71     | 128.90      | 19.17             |
|              | S2 (128 B)   | 4698.00        | 12132.12    | 94.78       | 26.95             |
|              | S3 (512 B)   | 13882.00       | 36009.97    | 70.33       | 36.75             |
|              | S4 (1024 B)  | 26104.00       | 67787.45    | 66.20       | 39.13             |
|              | S5 (4096 B)  | 99537.00       | 258708.05   | 63.16       | 41.13             |
|              | S6 (8192 B)  | 197509.00      | 513434.20   | 62.68       | 41.47             |
|              | S7 (16384 B) | 393159.00      | 1022123.10  | 62.39       | 41.67             |
|              | S8 (65536 B) | 1567795.00     | 4076169.65  | 62.20       | 41.80             |

## 5.3 Resource Consumption and Stability

| Instance     | Static memory (Bytes) | Peak memory (Bytes) | Stability |
|--------------|-----------------------|---------------------|-----------|
| Thunder-512  | 20480                 | 20480               | STABLE    |
| Thunder-768  | 20480                 | 20480               | STABLE    |
| Thunder-1024 | 20480                 | 20480               | STABLE    |

## 5.4 Transmission and Storage Overhead

Transmission and storage-overhead metrics for public keys, private keys, signatures, ciphertexts, key-exchange messages, and key-exchange rounds are not applicable to **Thunder**, because this submission is a cryptographic hash function and does not implement KEM, signature, or key-exchange functionality.

## 6 Raw Evidence and Submission Checklist

| Item                         | Evidence                                                                                                                                                      |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| KAT input directories        | Thunder_X86/API_CryptHash/Test_Vector/Thunder512,<br>Thunder_X86/API_CryptHash/Test_Vector/Thunder768,<br>Thunder_X86/API_CryptHash/Test_Vector/Thunder1024   |
| ngcc JSON result files       | Thunder512ref_X86_report.json.en,<br>Thunder768ref_X86_report.json.en,<br>Thunder1024ref_X86_report.json.en in<br>ThunderX86/ngcc_bench-main/build            |
| Commands and scripts         | testThunderX64.sh; manual command form documented in<br>ThunderX86/ngcc_bench-main/build/testThunderX64.sh                                                    |
| Dependency declaration       | Cryptographic Hash-Thunder-x86-Reference<br>Implementation Version/Dependency<br>Information/DEPENDENCIES.md; no third-party cryptographic<br>library is used |
| Material package<br>contents | Algorithm Source Code, Self-Assessment Code and Data,<br>and Dependency Information. This report is the<br>self-assessment report for the same package.       |

## 7 Final Confirmation

This report covers Thunder-512, Thunder-768, and Thunder-1024 for the Reference Implementation Version on the x86-64 Architecture. The reported functional, performance, memory, and stability values are derived from the raw ngcc evidence indexed above.